
Evaluating Traditional Machine Learning Models for Heart Disease Prediction

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OVERVIEW

The diagnosis of heart diseases is considered one of the most critical medical challenges due to the high prevalence worldwide and the severe complications resulting from delayed detection. With the rapid advancement of artificial intelligence techniques, the analysis of clinical data using machine learning algorithms has become an effective approach to support physicians in making accurate decisions based on quantitative and systematic data. These models provide exceptional capabilities in uncovering hidden patterns within medical data, making them promising tools for improving early diagnosis quality and reducing health risks associated with delayed detection.

Machine learning models for heart disease diagnosis rely on a set of essential clinical features that have been proven in medical literature to be closely associated with cardiac health, such as age, blood pressure, cholesterol levels, electrocardiogram (ECG) results, maximum heart rate during exertion, ST segment characteristics, and other vital indicators. The UCI Heart Disease dataset is one of the most widely used datasets in this field due to its clear features, ease of model implementation, and its extensive adoption in numerous academic studies.

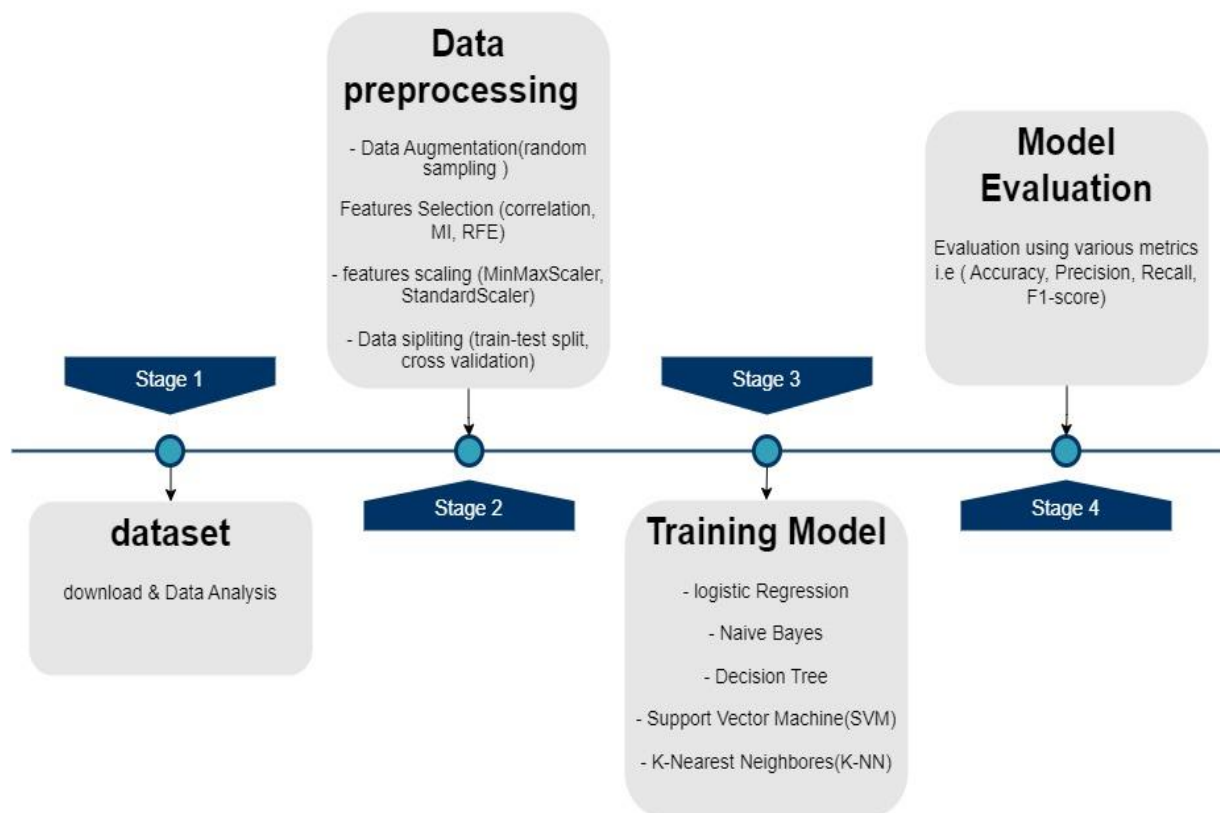
Among the recent studies addressing this topic is “Effectiveness of Machine Learning Models in Diagnosis of Heart Disease: A Comparative Study”, which presented a comparative analysis of several traditional machine learning algorithms, focusing on evaluating their effectiveness in diagnosing heart disease using basic clinical data.

Accordingly, this report aims to reassess the performance of traditional machine learning algorithms such as Logistic Regression, Naive Bayes, Decision Tree, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM) on the same dataset (UCI Heart Disease) and to conduct a direct comparison with the results reported in the aforementioned study.

The comparison process focuses on standard performance metrics such as Accuracy, Recall, Precision, and F1-Score, providing a comprehensive evaluation of each algorithm’s ability to predict the presence of the disease. This analysis also

helps identify the strengths and weaknesses of different models, particularly when dealing with sensitive and limited medical data, thereby enhancing the reliability of these models for potential clinical application.

Methodology



Dataset

URL [<https://www.kaggle.com/datasets/thisishusseinali/uci-heart-disease-data>]

Data Description

The dataset used here is the UCI Heart Disease Dataset, originating from the Cleveland Clinic Foundation, and is part of the well-known UCI Machine Learning Repository. The dataset contains 606 patient records, with each record representing an individual who underwent a series of medical and diagnostic examinations related to cardiovascular health.

The data include 14 core features, selected from a larger set of medical variables.

These features cover a wide range of physiological measurements, such as age, resting blood pressure, serum cholesterol level, fasting blood sugar, electrocardiogram (ECG) results, maximum heart rate achieved, exercise-induced angina, ST-segment depression induced by exercise, and the number of major vessels visible via fluoroscopy. These indicators are considered among the most important factors used by specialists in diagnosing heart diseases.

The primary task associated with this dataset is classification, i.e. predicting whether a patient has heart disease or not based on clinical features.

The target variable represents the presence of heart disease on a binary scale (0 = no disease, 1 = disease). This conversion facilitates the use of the dataset in classification experiments, enabling comparison of the performance of different algorithms.

Original Dataset Characteristics

- Clean dataset with no missing value.
- Categorical variables are already encoded.
- Samples showing the presence of the disease: 330 sample at rate 54.46%.
- Samples don't presence of the disease: 276 sample at rate 45.54%.

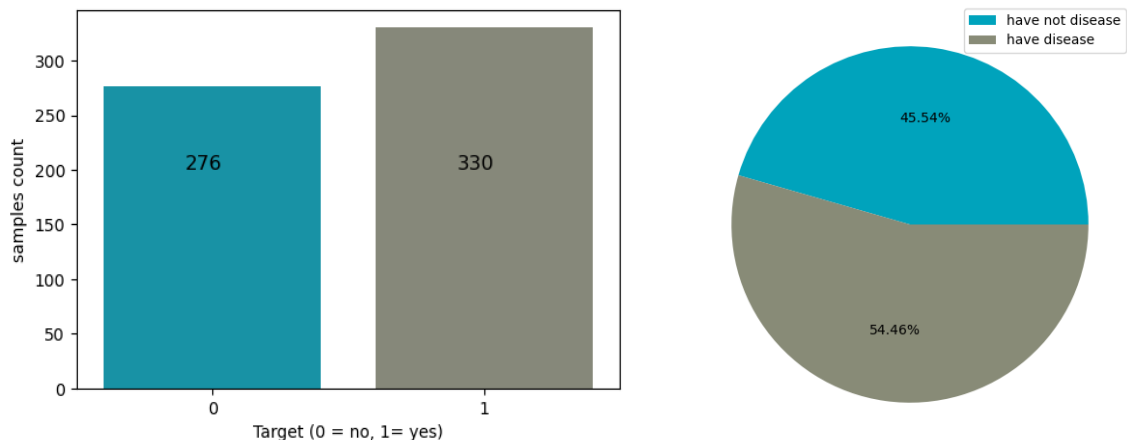


Fig. 1. Show sample count and rate

We notice a small degree of class imbalance in the data, which is minor and will be processed later to improve the model's accuracy.

Data Samples

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
383	67	0	2	152	277	0	1	172	0	0.0	2	1	2	1
282	51	1	3	125	213	0	0	125	1	1.4	2	1	2	1
201	67	0	0	106	223	0	1	142	0	0.3	2	2	2	1
578	59	1	0	164	176	1	0	90	0	1.0	1	2	1	0
284	44	1	2	120	226	0	1	169	0	0.0	2	0	2	1

The dataset contains 14 features, and the following table presents these features with a brief description of each.

Table 1. Details of the features in the data set

Feature Name	Description
age	The person's age in years
sex	The person's sex (0 = female, 1 = male)
cp	The chest pain experienced (value 0: typical angina, value 1: atypical angina, value 2: non-anginal pain, value 4: asymptomatic)
trestbps	The person's resting blood pressure (mm Hg on admission to the hospital)
chol	The person's cholesterol measurement in mg/dl
fbs	The person's fasting blood sugar (> 120 mg/dl, 1 = true; 0 = false)
restecg	Resting electrocardiographic measurement (0 = normal, 1 = having ST-T wave abnormality, 2 = showing probable or definite left ventricular hypertrophy by Estes' criteria)
thalach	The person's maximum heart rate achieved
exang	Exercise induced angina (1 = yes; 0 = no)
oldpeak	ST depression induced by exercise relative to rest ('ST' relates to positions on the ECG plot.)
slope	the slope of the peak exercise ST segment (value 0: upsloping, value 1: flat, value 2: down sloping)
ca	The number of major vessels (0-3)
thal	A blood disorder called thalassemia (0 = null; 1 = fixed defect; 2 = normal; 3 = reversable defect)
target	Heart disease (0 = no, 1 = yes)

Data Preprocessing

Data Augmentation

Data Augmentation refers to a set of techniques aimed at artificially increasing the size and diversity of a dataset. It is commonly used to address issues such as class imbalance, limited data, and overfitting, thereby improving the model's generalization ability. One of the simplest augmentation strategies for tabular dataset is Random Sampling (Over-sampling), which duplicates samples from the minority class to achieve a more balanced dataset distribution.

- Positive cases (heart disease): 330 cases.
- Negative cases (no heart disease): 276 cases.

After applying Over-sampling:

- Final Dataset size for each class: around 660 records per class.
- Purpose: to achieve a relative balance between classes, enhancing model performance and reducing bias.

Features Selection

Three Techniques were applied:

- **Correlation:** Calculate the linear relationship between each feature and the target variable using corr. coefficients (Pearson). Feature with high `corr. with the target` are consider more useful, while low-Corr features can be ignored to reduce complexity and noise in the data. Top Features: [cp, thalach, slope, exang, oldpeak, ca, thal]
- **Mutual Information (MI):** Measure the amount of info. shared between each feature and the target variable. In other words, it shows how well a feature can predict the class. The feature with high MI is consider more important for the model. Top Features: [chol :0.264973, cp: 0.183739, thalach: 0.171626, oldpeak: 0.129939, thal: 0.113173, ca: 0.104502, exang: 0.104303]
- **Recursive Feature Elimination (RFE):** Select the most important features recursively by fitting a model and removing the least important features step

by step until the optimal set of features is obtained. Top Features: [sex, cp, exang, oldpeak, slope, ca, thal]

Data Scaling and Splitting

Two scaling methods were tested:

- Min-Max Normalization (MinMaxScaler)
- Standardization (StandardScaler)

Models were trained twice: once on normalized data and once on standardized data. StandardScaler consistently gave higher performance for most algorithms.

Data Splitting

To ensure a fair and reliable evaluation of the model, two different data-splitting strategies were applied. Each method provides a unique perspective on the model's performance.

- **Train-test split:** The dataset was first divided using the standard train-test split approach, where a fixed portion of the data is used for training and the remaining portion for testing.
 - **Training set: 80%**
 - **Testing set: 20%**
 - **Purpose:** This method provides an initial straightforward evaluation of the model on unseen data.
- **K-Fold Cross Validation:** A more robust evaluation was conducted using K-Fold Cross Validation, where the data is repeatedly split into multiple subsets(fold).
 - **No. of folds (cv): 9**
 - **Procedure:** The dataset is divided into 9 equal folds. In each iteration, the model is trained on 8 folds and tested on the remaining fold. This process is repeated until every fold has served as a test set once.
 - **Evolution:** Average performance across all folds.
 - **Purpose:** Provides a more stable and reliable estimate of model performance, reducing the bias that may occur from a single data split.

Using both splitting strategies offers a more comprehensive evaluation:

- Train-Test split gives a quick performance snapshot.
- K-Fold Cross Validation ensure the model performs consistently across different subsets of the data.

Trained Models and Performance Result

In this section, several machine learning algorithms were trained and evaluated using four performance metrics: Accuracy, Precision, Recall, and F1-Score. Five different classifiers were used: Logistic Regression (LR), Nive Bayes (NB), Decision Tree (DT), Super Vector Machine (SVM), K-Nearest Neighbor (K-NN), and each model was trained under different preprocessing and feature-selection settings to ensure a comprehensive performance comparison.

Feature Selection via Correlation Features

To understand the relationships between the features in the dataset and identify which variables may have a stronger influence on the target class, a Correlation Analysis was performed. This step is essential before training the models, as it helps in selecting the most relevant features, reducing noise, and improving model performance.

Purpose of Correlation Analysis

- Identify highly correlated features that may contribute strongly to the prediction.
- Detect redundant features that show extremely high correlation with each other.
- Determine the strength and direction of relationships (positive or negative) between variables.
- Provide insights that guide Feature Selection before model training.

Correlation Matrix

A Correlation Matrix was generated to visualize the statistical relationships between all numerical features.

The matrix displays correlation values ranging from -1 to +1, where:

+1 = perfect positive correlation, -1 = perfect negative correlation
0 = no correlation.



Fig. 2. Correlation Features analysis using Pearson correlation.

To provide a clearer visual understanding of feature relationships, a bar plot of correlation values was generated. This visualization makes it easier to quickly identify the most influential features at a glance, helping simplify the feature-selection process.

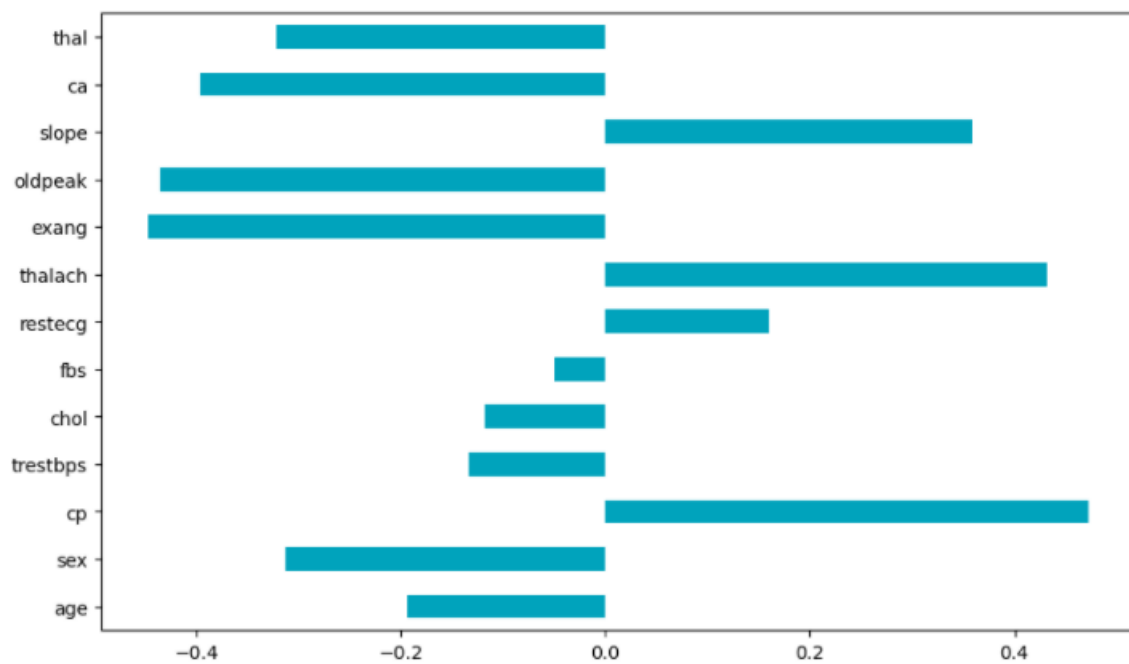


Fig. 3. Bar chart of correlation features with the target

Based on the correlations observed in the correlation matrix and the bar plot, [cp, thalach, slope, exang, oldpeak, ca, thal] were selected as the most relevant for predicting the target variable. These features showed the strongest relationships with the target and were therefore chosen to train the models in this section.

The machine learning models presented in this section were trained using only these seven features, as they were identified through the correlation analysis to have the highest predictive value.

For each classifier, performance was assessed using both Standardized and Normalized data to determine which preprocessing approach yielded better results

Logistic Regression

Logistic Regression is a linear model that predicts the probability of a binary outcome using the logistic (sigmoid) function. It works by estimating coefficient for each feature to model their influence on the target. Its simplicity makes it fast, interpretable, and effective for linearly separable data.

Training Result:

Table 2. LR Performance: Standardized vs Normalized Data

Preprocessing	Accuracy	Precision	Recall	F1-Score
Normalization	81.82	78.79	83.87	81.25
Standardization	85.61	84.13	85.48	84.8

Naïve Bayes

Naïve Bayes is a probability classifier that applies Bayes' theorem assuming feature independence. It estimates the probability of each class given the feature values. NB is simple, extremely fast, works well with small datasets, and is robust to irrelevant features.

Training Result:

Table 3. NB Performance: Standardized vs Normalized Data

Preprocessing	Accuracy	Precision	Recall	F1-Score
Normalization	77.27	76.67	74.19	75.41
Standardization	77.27	76.67	74.19	75.41

Decision Tree

Decision Tree splits the data recursively based on feature thresholds to maximize purity in child nodes. It builds an interpretable tree structure representing decision rules. DT captures non-linear relationships, handles both numerical and categorical features, and is easy to interpret, though it can overfit if not properly constrained.

Training Result:

Table 4. DT Performance: Standardized vs Normalized Data

Preprocessing	Accuracy	Precision	Recall	F1-Score
Normalization	93.18	89.55	96.77	93.02
Standardization	93.18	89.55	96.77	93.02

Support Vector Machine

SVM finds the optimal hyperplane that maximize the margin between classes. Using kernels, it can handle non-linear separation. SVM is effective in high-dimensional space, robust to overfitting, and captures both linear and non-linear relationships.

Training Result:

Table 5. SVM Performance: Standardized vs Normalized Data

Preprocessing	Accuracy	Precision	Recall	F1-Score
Normalization	89.39	86.36	91.94	89.06
Standardization	90.15	88.89	90.32	89.6

K-Nearest Neighbor

KNN classifies a sample based on the majority label among its k nearest neighbors in feature space. It operates without an explicit training phase and relies on distance metrics to measure similarity. KNN can capture complex non-linear patterns and adapts to local data structures.

Training Result:

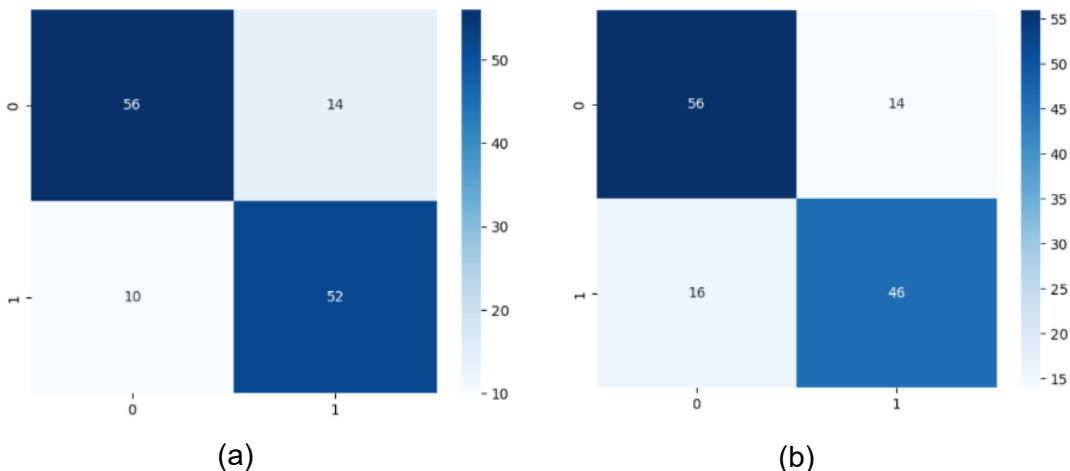
Table 6. KNN Performance: Standardized vs Normalized Data

Preprocessing	Accuracy	Precision	Recall	F1-Score
Normalization	87.12	82.61	91.94	87.02
Standardization	83.33	81.25	83.87	82.54

Performance Result Comparison

Standardized vs Normalized

In the following analysis, confusion metrics for the five models are presented using tow types of data preprocessing: data after applying Normalization and data after applying Standardization. This comparison aims to illustrate the impact of each preprocessing method on model performance, where each plot shows the number of correctly and incorrectly classified instances per class, facilitating the evaluation of how model accuracy improves or declines when the data scaling is changed.



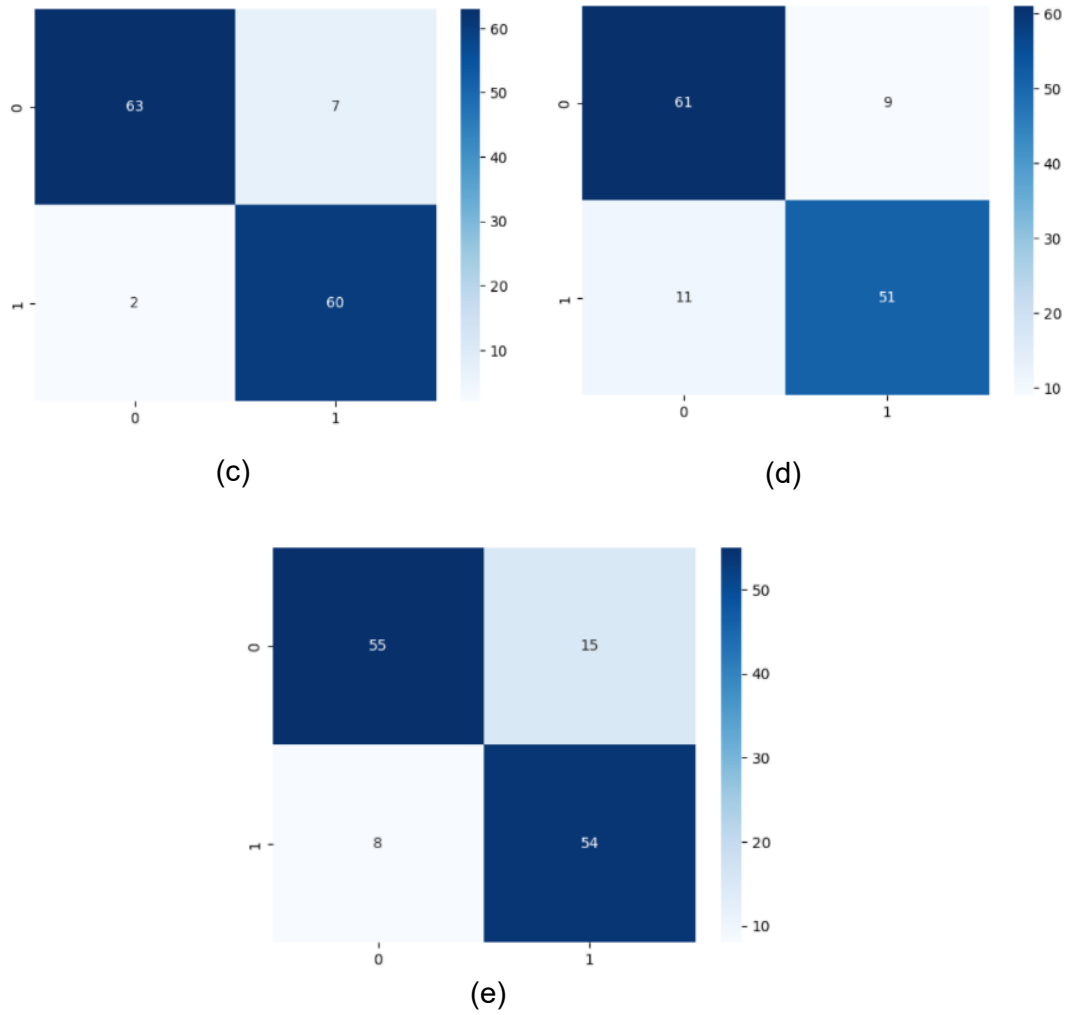


Fig. 4. Confusion matrix obtained from (a)LR, (b)NB, (c)DT, (d)SVM, (e)KNN with Normalized data

Table 7. Compare all model's performance with Normalized data

Model Name	Accuracy	Precision	Recall	F1-Score
Logistic Regression	81.82	78.79	83.87	81.25
Naïve Bayes	76.67	74.19	75.41	75.41
Decision Tree	93.18	89.55	96.77	93.02
Support Vector Machin	84.85	85.00	82.26	83.61
K-Nearest Neighbor	82.58	78.26	87.10	82.44

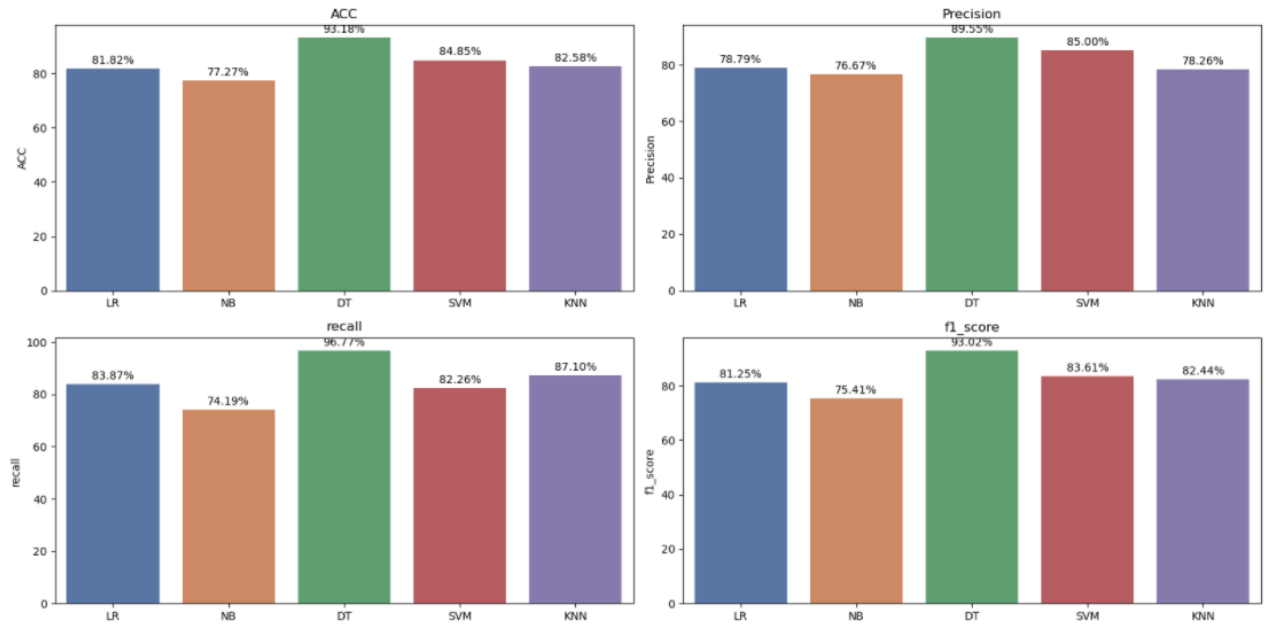
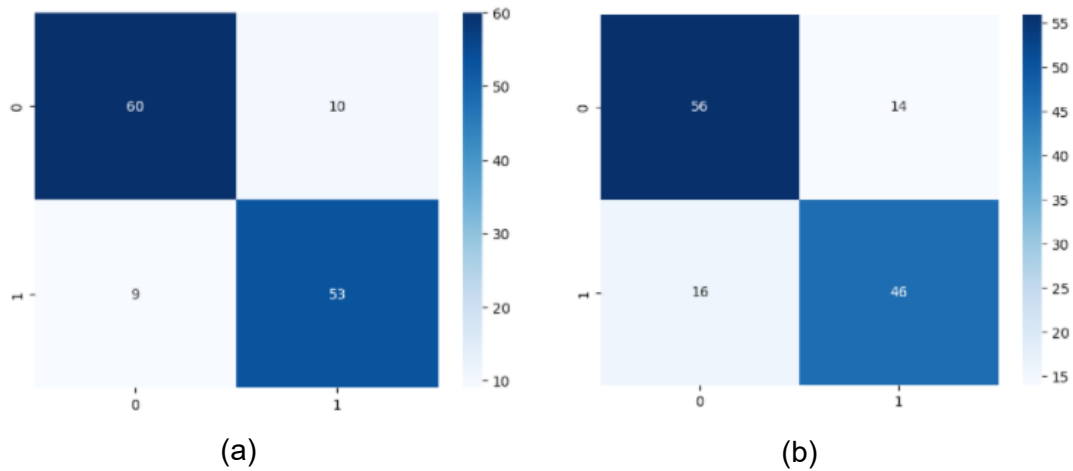


Fig. 5. Result obtained from the ML models with normalized data and features selected by its correlation, Accuracy, Precision, Recall, F1-score



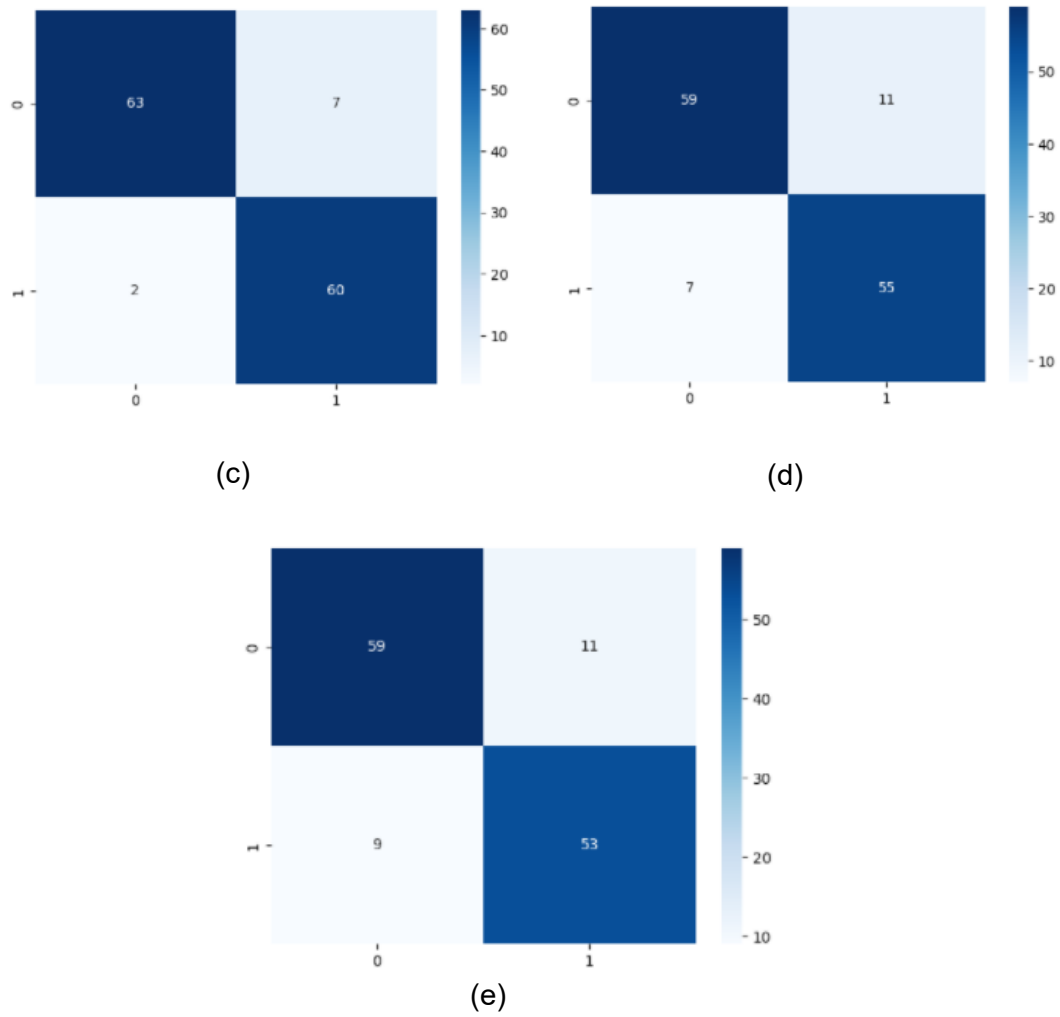


Fig. 6. Confusion matrix obtained from (a)LR, (b)NB, (c)DT, (d)SVM, (e)KNN with Standardized data

Table 8. Compare All Model's Performance with Standardization data

Model Name	Accuracy	Precision	Recall	F1-Score
Logistic Regression	85.61	84.13	85.48	84.80
Naïve Bayes	77.27	76.67	74.19	75.41
Decision Tree	93.18	89.55	96.77	93.02
Support Vector Machin	86.36	83.33	88.71	85.94
K-Nearest Neighbor	84.85	82.81	85.48	84.13

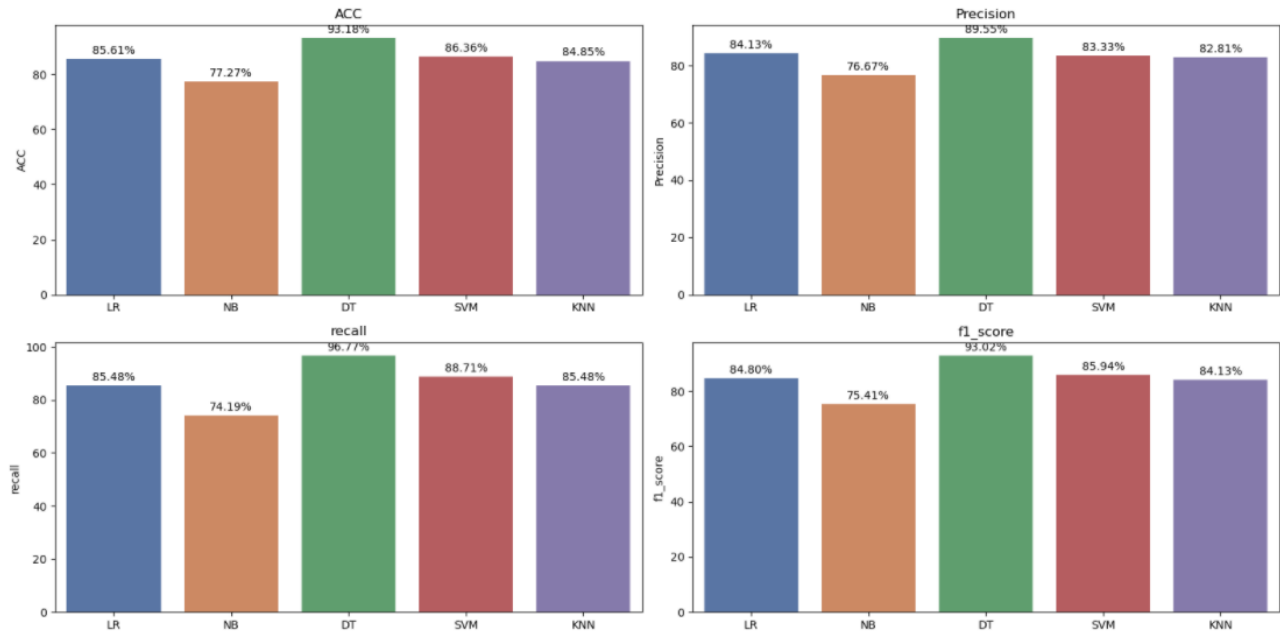


Fig. 7. Result obtained from the ML models with Standardized data and features selected by its correlation, Accuracy, Precision, Recall, F1-score.

Upon comparing the performance metrics, it was generally observed that models trained on standardized data achieved similar or slightly higher results compared to those trained on normalized data. An exception was noted for the K-Nearest Neighbors (KNN) classifier, where normalization resulted in higher accuracy. Based on these observations, cross-validation experiments were conducted using standardized data along with feature selection methods (Correlation, MI, and FRE) to ensure a more consistent and reliable evaluation of model performance and to enable comparison of results across methods.

Cross- Validation Performance Using Standardized Data

In this section, cross-validation is applied to evaluate the robustness and generalization performance of the models. Unlike the previous experiment, which compared standardized and normalized data using a simple train-test split, the cross-validation analysis is conducted exclusively on standardized data. This choice was based on earlier observations indicating that standardization consistently produced equal or superior results.

To further examine the effectiveness of feature selection, features correlation, Recursive Feature Elimination, Mutual Information are applied. The models are trained and evaluated under each method using K-fold Cross-Validation, allowing for a reliable comparison of how each feature selection technique influences model performance.

The following tables and plots present the cross-validation performance results for all models using the previously discussed features selection techniques.

Table 9. CV Metrics – correlation features technique

Model Name	Accuracy	Precision	Recall	F1-Score
Logistic Regression	83.81	84.44	83.32	83.39
Naïve Bayes	82.00	82.22	81.66	81.71
Decision Tree	94.73	94.86	94.63	94.69
Support Vector Machin	87.61	88.12	87.18	87.35
K-Nearest Neighbor	86.12	86.33	85.74	85.88

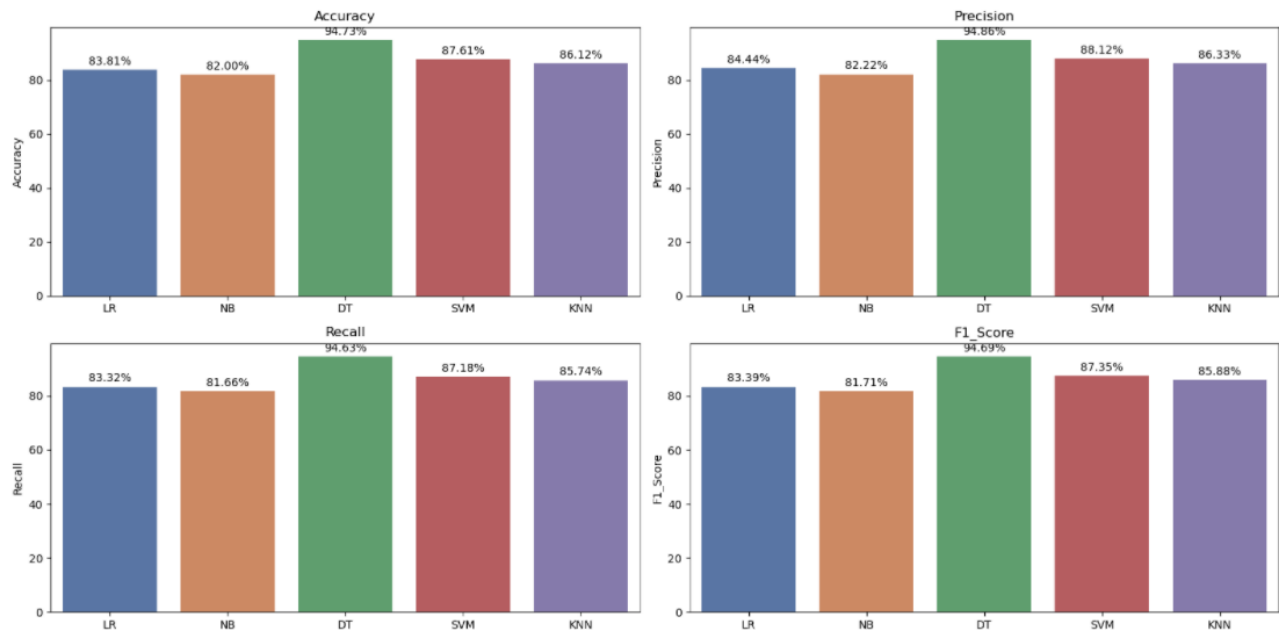


Fig. 8. CV performance comparison – correlation features technique

Table 10. CV Metrics – ML technique

Model Name	Accuracy	Precision	Recall	F1-Score
Logistic Regression	82.49	83.13	81.91	82.03
Naïve Bayes	83.15	83.43	82.78	82.85
Decision Tree	92.07	92.18	91.95	91.99
Support Vector Machin	85.30	85.63	84.84	85.00
K-Nearest Neighbor	84.64	85.20	84.13	84.29

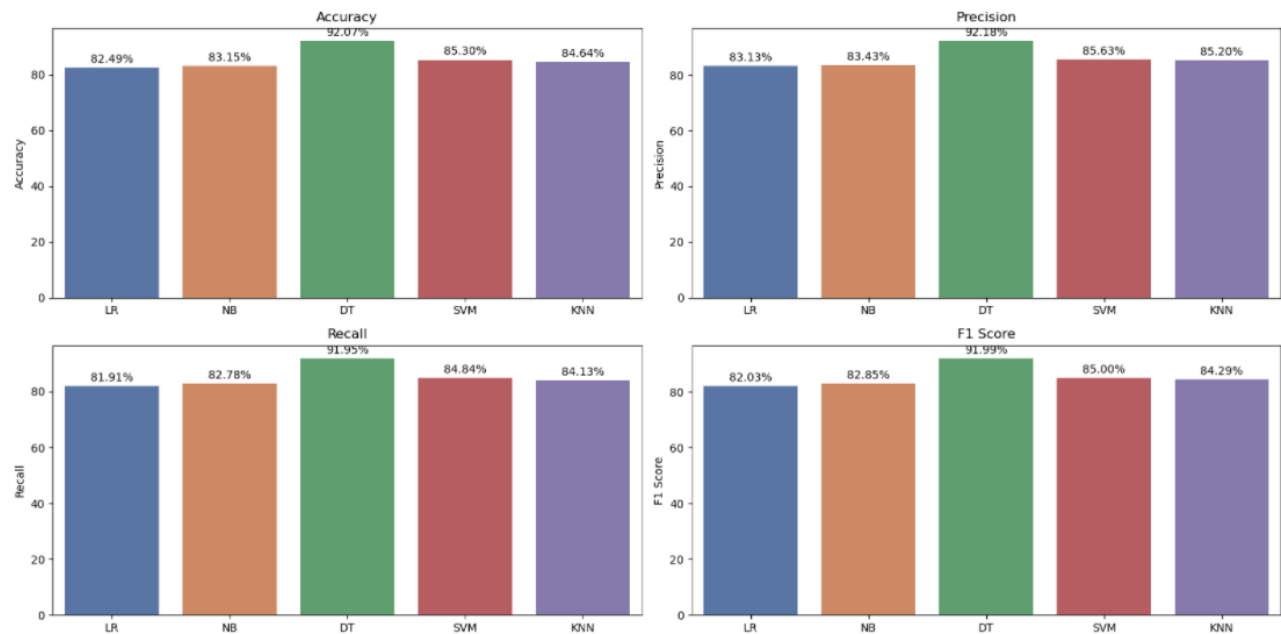
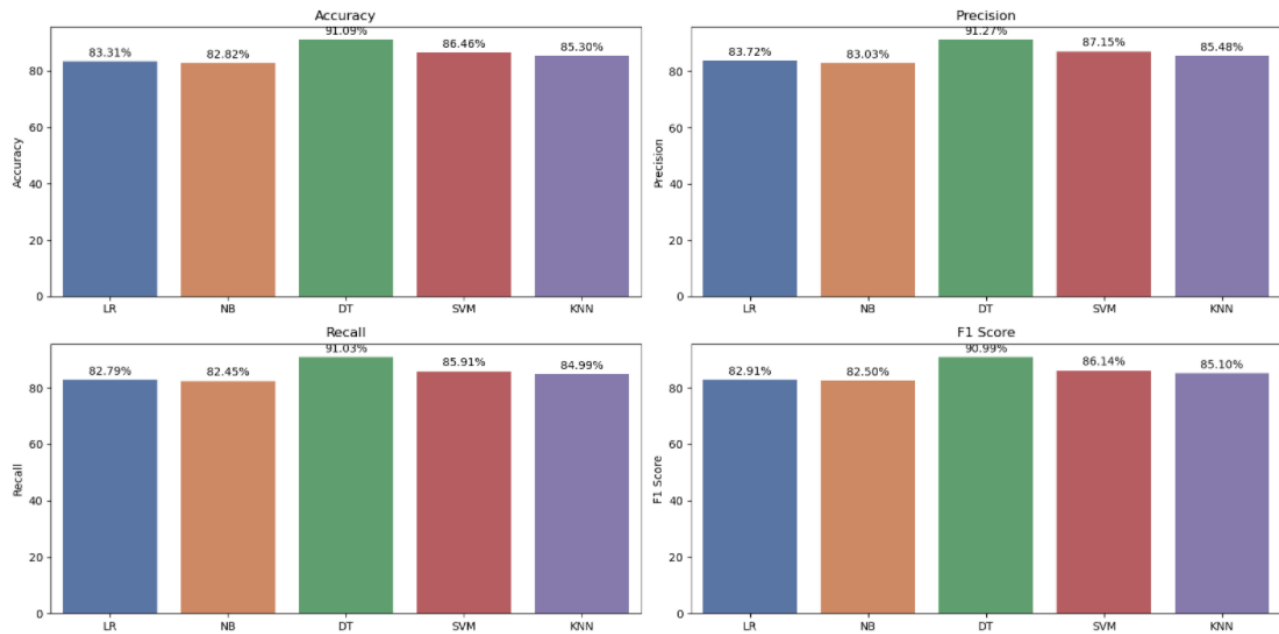


Fig. 9. CV performance comparison – ML technique

Table 11. CV Metrics – RFE technique

Model Name	Accuracy	Precision	Recall	F1-Score
Logistic Regression	83.31	83.72	82.79	82.91
Naïve Bayes	82.82	83.03	82.45	82.50
Decision Tree	91.09	91.27	91.03	90.99
Support Vector Machin	86.46	87.15	85.91	86.14
K-Nearest Neighbor	85.30	85.48	84.99	85.10

**Fig. 10.** CV performance comparison – RFE technique

Discussion

Based on the comprehensive evaluation conducted using both the tarin-test split and cross-validation techniques, the results consistently showed that the Decision Tree classifier achieved the highest performance among all models. Across all feature selection method Correlation, MI, and RFE the Decision Tree maintained exceptional accuracy, precision, recall, and f1-score, often exceeding 91% and reaching up to 94.73%, which highlights its strong ability to capture the underlying patterns in the dataset.

In contrast, classifiers such as Logistic Regression, Naïve Bayes, SVM, and K-NN demonstrated moderate and more stable performance, with accuracy scores generally ranging between 82% and 87% depending on the preprocessing methods and feature selection technique. These models showed consistent behavior but did not match the high predictive power demonstrated by the Decision Tree.

Moreover, the Cross-Validation results confirmed the superiority of the Decision Tree, showing that its high accuracy was not due to data-splitting bias but reflected genuine robustness and generalization capability. Additionally, the feature selection techniques produced consist rankings where key features such as cp, thalach, oldpeak, slope, exang, ca, and thal repeatedly contributed to improving the model's predictive power.

During the experiments, the Decision Tree Model initially showed very high accuracy values on the training data, indicating potential overfitting. By plotting the model performance against different tree depths, an optimal depth was selected that balances training and validation accuracy. This approach prevents overfitting while maintaining high predictive performance. The higher value observed for Decision Tree

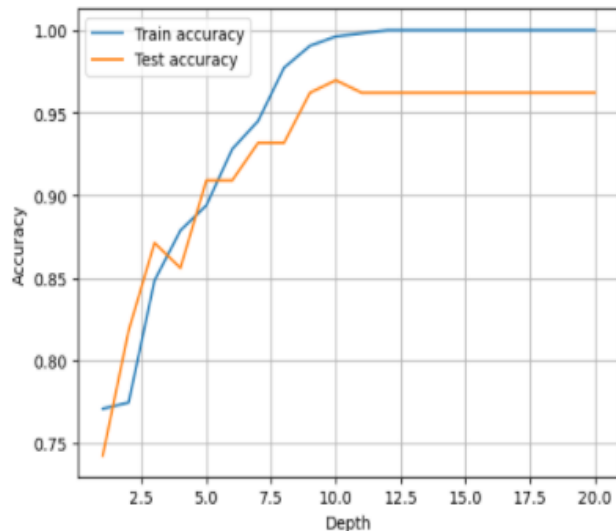


Fig. 11. Train-Test Acc Compare

compared other algorithms is due to its ability to capture complex non-linear relationships in the data. In comparison with the study conducted by Waleed Alsabhan and Abdullah Alfadhly, which focused on advanced boosting models such as GBM, CatBoost, and LightGBM, our result demonstrates notable difference. While their highest reported accuracy reached 90.2% using GBM under min-max scaling, the Decision Tree model in our study achieved a significantly higher accuracy exceeding 94.73%. This superior performance is primarily attributed to the effective application of feature selection techniques, which enabled the model to

focus on the most informative features. Unlike the previous study, which emphasized advanced ensemble algorithms, our approach shows that even traditional classifiers can outperform